MILITARY SPECIFICATION SHEET

# WIRE, ELECTRIC, FLUOROPOLYMER-INSULATED, EXTRUDED TFE, SILVER-COATED COPPER CONDUCTOR, 1000-VOLT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of Specification MIL-W-22759.

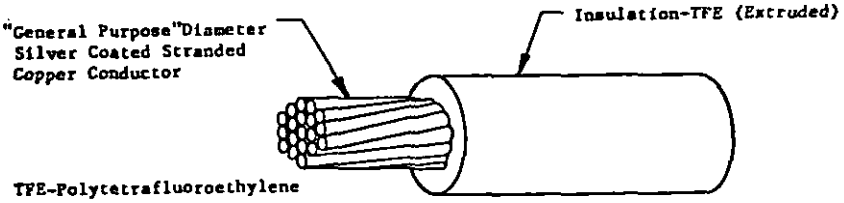



TABLE I. CONSTRUCTION DETAILS

| Part No. 1/ | Wire size | Stranding (Number of strands X AWG gage of strands) | Diameter of stranded conductor (inches) | | Finished wire | | |
|---|---|---|---|---|---|---|---|
| | | | (min) | (max) | Resistance at 20°C (68°F) (ohms/1000 ft) (max) | Diameter (inches) | Weight (lbs/1000 ft) (max) |
| M22759/9-28-* | 28 | 7 X 36 | .014 | .015 | 63.8 | .043 ±.002 | 1.90 |
| M22759/9-26-* | 26 | 19 X 38 | .018 | .020 | 38.4 | .048 ±.002 | 2.57 |
| M22759/9-24-* | 24 | 19 X 36 | .023 | .025 | 24.3 | .053 ±.002 | 3.33 |
| M22759/9-22-* | 22 | 19 X 34 | .029 | .032 | 15.1 | .060 ±.002 | 4.60 |
| M22759/9-20-* | 20 | 19 X 32 | .037 | .040 | 9.19 | .068 ±.002 | 6.40 |
| M22759/9-18-* | 18 | 19 X 30 | .046 | .050 | 5.79 | .078 ±.002 | 9.10 |
| M22759/9-16-* | 16 | 19 X 29 | .052 | .057 | 4.52 | .085 ±.002 | 11.0 |
| M22759/9-14-* | 14 | 19 X 27 | .065 | .072 | 2.88 | .100 ±.003 | 16.4 |
| M22759/9-12-* | 12 | (B) 19 X 25 | (B) .082 | (B) .090 | (B) 1.81 | .120 ±.004 | (B) 25.3 |
| M22759/9-10-* | 10 | 37 X 26 | .106 | .112 | 1.19 | .141 ±.004 | 38.2 |
| M22759/9-8-* | 8 | 133 X 29 | .158 | .169 | .658 | .207 ±.005 | 68.8 |

1/ PART NO.: The asterisks in the part number column, Tables I through III, shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white - M22759/9-20-9; white with orange stripe - M22759/9-20-93.

(B) denotes changes



FSC 6145

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TABLE II. PERFORMANCE DETAILS

| Part No. | Abrasion resistance (Procedure 11) | | | | Bend testing | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Mandrel diameter (inches) (±3%) | | Test load (lbs) (±3%) | |
| | Resistance (inches of tape) (min) (initial condition) | Weight support bracket | Weight (lbs) | Tension load (lbs) | Life cycle (oven & bend tests) 1/ | Cold bend test | Life cycle (oven & bend tests) 1/ | Cold bend test |
| M22759/9-28-* | 24 | A | .50 | 1.0 | .125 | .250 | .50 | |
| M22759/9-26-* | 24 | A | .50 | 1.0 | .125 | .250 | .50 | |
| M22759/9-24-* | 30 | A | .50 | 1.0 | .125 | .250 | .50 | |
| M22759/9-22-* | 30 | A | .50 | 1.0 | .250 | .375 | .75 | |
| M22759/9-20-* | 30 | A | .50 | 1.0 | .250 | .375 | .75 | |
| M22759/9-18-* | 36 | B | .75 | 1.0 | .250 | .375 | 1.00 | |
| M22759/9-16-* | 36 | B | .75 | 2.0 | .375 | .500 | 1.00 | |
| M22759/9-14-* | 36 | B | 1.25 | 2.0 | .500 | .750 | 2.00 | |
| M22759/9-12-* | 36 | B | 1.25 | 2.0 | .500 | .750 | 2.00 | |
| M22759/9-10-* | 48 | B | 1.25 | 2.0 | .750 | 1.00 | 3.00 | |
| M22759/9-8-* | 48 | C | 2.00 | 2.0 | .750 | 1.00 | 3.00 | |

1/ Also for bend tests after immersion.

## ADDITIONAL REQUIREMENTS

TEMPERATURE RATING: 200°C (392°F) max conductor temperature
VOLTAGE RATING: 1000 volts (rms) at sea level
SPARK TEST OF PRIMARY INSULATION: Not required
IMPULSE DIELECTRIC TEST: 9.5 kilovolts (peak), 100% test
INSULATION RESISTANCE: 50,000 megohms for 1000 ft (min)
WRAP TEST: "Wrap back" test required, no cracking
Oven temperature, 313 ±2°C (595.4 ±3.6°F)
BLOCKING: 260 ±2°C (500 ±3.6°F)
SHRINKAGE: 0.03 inch max at 290 ±2°C (554 ±3.6°F)
WICKING: No requirement
LOW TEMPERATURE (COLD BEND):
Bend temperature: -65 ±2°C (-85 ±3.6°F)
Dielectric test, 3000 volts (rms), 60Hz
THERMAL SHOCK: Oven temperature 200 ±2°C (392 ±3.6°F)
Max change in measurement Sizes 28 through 12: 0.060 inch
Sizes 10 through 8: 0.100 inch
FLAMMABILITY: Post-flame dielectric test not required
LIFE CYCLE: Oven temperature 275 ±2°C (527 ±3.6°F)
Dielectric test, 3000 volts (rms), 60Hz
DIELECTRIC TEST AFTER IMMERSION: 3000 volts (rms), 60Hz
ABRASION RESISTANCE AFTER IMMERSION: No requirement
HUMIDITY RESISTANCE: No requirement
SURFACE RESISTANCE: 500 megohm-inches (min) initial and final readings
SMOKE: 290°C (554°F)
ACID RESISTANCE: Required
Dielectric test, 3000 volts (rms), 60Hz
COLOR: In accordance with MIL-STD-104, Class 1; white preferred

COLOR STRIPING OR BANDING DURABILITY: 250 cycles (500 strokes) (min), 500 grams weight
IDENTIFICATION DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
WIRE LENGTH REQUIREMENTS: Schedule A



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SUPERSESSION DATA: The wire of this specification sheet, by part number, replaces and supersedes the wire of MS18113(AS) (canceled) in accordance with Table III.

TABLE III. SUPERSESSION BY PART NUMBER

| Part number MS18113(AS) | Part number MIL-W-22759/9 |
|---|---|
| MS18113-28-* | M22759/9-28-* |
| MS18113-26-* | M22759/9-26-* |
| MS18113-24-* | M22759/9-24-* |
| MS18113-22-* | M22759/9-22-* |
| MS18113-20-* | M22759/9-20-* |
| MS18113-18-* | M22759/9-18-* |
| MS18113-16-* | M22759/9-16-* |
| MS18113-14-* | M22759/9-14-* |
| MS18113-12-* | M22759/9-12-* |
| MS18113-10-* | M22759/9-10-* |
| MS18113-8-* | M22759/9-8-* |

Custodians:
Navy - AS
Army - EL
Air Force - 80

Preparing activity:
Navy - AS
(Project No. 6145-0670-7)

Review activities:
Navy - EC, OS
Army - ME, MI, MU
Air Force -
DSA - IS
NSA

User activities:
Navy - MC
Army - AT, AV
Air Force - 11

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.



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DOCUMENT IDENTIFIER AND TITLE MIL-W-22759/8B WIRE, ELECTRIC, FLUOROPOLYMER-INSULATED, ABRASION RESISTANT, EXTRUDED TFE, MEDIUM WEIGHT, NICKEL-COATED COPPER CONDUCTOR, 600-VOLT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

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